

```

library(data.table)
library(ggplot2)
library(lubridate)

filePath <- "C:/Users/tracy/Downloads/QVI_data.csv"
data <- fread(filePath)

data[, DATE := as.Date(DATE, format="%Y-%m-%d")]
data[, YEARMONTH := year(DATE)*100 + month(DATE)]

monthly_metrics <- data[, .(
  total_sales = sum(TOT_SALES, na.rm = TRUE),
  n_customers = uniqueN(LYLT_Y_CARD_NBR),
  avg_txn_per_customer = .N / uniqueN(LYLT_Y_CARD_NBR)
), by = .(STORE_NBR, YEARMONTH)]

pre_trial_end <- 201901
stores_full_obs <- monthly_metrics[YEARMONTH <= pre_trial_end, .N, by = STORE_NBR]
[N == length(unique(monthly_metrics$YEARMONTH[monthly_metrics$YEARMONTH <=
pre_trial_end])), STORE_NBR]
pre_trial_metrics <- monthly_metrics[STORE_NBR %in% stores_full_obs & YEARMONTH <=
pre_trial_end]

calculateCorrelation <- function(inputTable, metricCol, trial_store) {
  calcCorrTable <- data.table(Store1 = numeric(), Store2 = numeric(), corr_measure
= numeric())
  storeNumbers <- unique(inputTable$STORE_NBR[inputTable$STORE_NBR !=
storeComparison])

  for (i in STORE_NBR) {
    trial_values <- inputTable[STORE_NBR == storeComparison, get(metricCol)]
    control_values <- inputTable[STORE_NBR == i, get(metricCol)]
    corr_val <- cor(trial_values, control_values, use = "complete.obs")

    calcCorrTable <- rbind(calcCorrTable, data.table(
      Store1 = storeComparison,
      Store2 = i,
      corr_measure = corr_val
    ))
  }
  return(calcCorrTable)
}

calculateMagnitudeDistance <- function(inputTable, metricCol, trial_store) {
  store_numbers <- setdiff(unique(inputTable$STORE_NBR), trial_store)
  result <- data.table(Store1 = integer(), Store2 = integer(), mag_measure =
numeric())
  for (store in store_numbers) {
    diff_vec <- abs(inputTable[STORE_NBR == trial_store, get(metricCol)] -
inputTable[STORE_NBR == store, get(metricCol)])
    mag <- 1 - (diff_vec - min(diff_vec)) / (max(diff_vec) - min(diff_vec))
    result <- rbind(result, data.table(Store1 = trial_store, Store2 = store,
mag_measure = mean(mag)))
  }
  return(result)
}

trial_stores <- c(77, 86, 88)
trial_period <- 201902:201904

```

```

for (trial_store in trial_stores) {
  corr_sales <- calculateCorrelation(pre_trial_metrics, "total_sales", trial_store)
  corr_customers <- calculateCorrelation(pre_trial_metrics, "n_customers",
trial_store)
  mag_sales <- calculateMagnitudeDistance(pre_trial_metrics, "total_sales",
trial_store)
  mag_customers <- calculateMagnitudeDistance(pre_trial_metrics, "n_customers",
trial_store)

  score_sales <- merge(corr_sales, mag_sales, by = c("Store1", "Store2"))[,
score_sales := 0.5*corr_measure + 0.5*mag_measure]
  score_customers <- merge(corr_customers, mag_customers, by =
c("Store1", "Store2"))[, score_customers := 0.5*corr_measure + 0.5*mag_measure]

  score_control <- merge(score_sales[, .(Store1, Store2, score_sales)],
score_customers[, .(Store1, Store2, score_customers)], by = c("Store1", "Store2"))
  score_control[, final_score := 0.5*score_sales + 0.5*score_customers]

  control_store <- score_control[order(-final_score)][1, Store2]
  cat("Trial Store:", trial_store, "- Control Store:", control_store, "\n")

  metrics_viz <- monthly_metrics[STORE_NBR %in% c(trial_store, control_store)]
  metrics_viz[, Store_type := ifelse(STORE_NBR == trial_store, "Trial", "Control")]

  scaling_sales <- sum(pre_trial_metrics[STORE_NBR == trial_store, total_sales]) /
sum(pre_trial_metrics[STORE_NBR == control_store, total_sales])
  scaling_customers <- sum(pre_trial_metrics[STORE_NBR == trial_store,
n_customers]) / sum(pre_trial_metrics[STORE_NBR == control_store, n_customers])
  metrics_viz[STORE_NBR == control_store, total_sales := total_sales *
scaling_sales]
  metrics_viz[STORE_NBR == control_store, n_customers := n_customers *
scaling_customers]

  ggplot(metrics_viz[YEARMONTH <= pre_trial_end], aes(x = YEARMONTH, y =
total_sales, color = Store_type)) +
    geom_line() +
    labs(title = paste("Pre-trial Total Sales: Trial store", trial_store))

  ggplot(metrics_viz[YEARMONTH <= pre_trial_end], aes(x = YEARMONTH, y =
n_customers, color = Store_type)) +
    geom_line() +
    labs(title = paste("Pre-trial Customers: Trial store", trial_store))

  pct_diff_sales <- merge(
    metrics_viz[STORE_NBR == control_store & YEARMONTH %in% trial_period, .
(YEARMONTH, control_sales = total_sales)],
    metrics_viz[STORE_NBR == trial_store & YEARMONTH %in% trial_period, .
(YEARMONTH, trial_sales = total_sales)],
    by = "YEARMONTH"
  )
  pct_diff_sales[, pct_diff := (trial_sales - control_sales) / control_sales]

  pct_diff_customers <- merge(
    metrics_viz[STORE_NBR == control_store & YEARMONTH %in% trial_period, .
(YEARMONTH, control_customers = n_customers)],
    metrics_viz[STORE_NBR == trial_store & YEARMONTH %in% trial_period, .
(YEARMONTH, trial_customers = n_customers)],
    by = "YEARMONTH"
  )

```

```

)
pct_diff_customers[, pct_diff := (trial_customers - control_customers) /
control_customers]

cat("Percentage differences (sales) for Trial Store", trial_store, ":\n")
print(pct_diff_sales)
cat("Percentage differences (customers) for Trial Store", trial_store, ":\n")
print(pct_diff_customers)

pre_trial_sales <- metrics_viz[STORE_NBR %in% c(trial_store, control_store) &
YEARMONTH <= pre_trial_end]
std_sales <- sd((pre_trial_sales[Store_type=="Trial", total_sales] -
pre_trial_sales[Store_type=="Control",
total_sales])/pre_trial_sales[Store_type=="Control", total_sales])
std_customers <- sd((pre_trial_sales[Store_type=="Trial", n_customers] -
pre_trial_sales[Store_type=="Control",
n_customers])/pre_trial_sales[Store_type=="Control", n_customers])

ggplot(metrics_viz[YEARMONTH %in% trial_period], aes(x = YEARMONTH)) +
  geom_line(aes(y = total_sales, color = Store_type)) +
  geom_ribbon(data = metrics_viz[STORE_NBR == control_store & YEARMONTH %in%
trial_period],
            aes(ymin = total_sales*(1-2*std_sales), ymax =
total_sales*(1+2*std_sales)),
            fill = "grey80", alpha = 0.3) +
  labs(title = paste("Trial Sales vs Control Store", trial_store))

ggplot(metrics_viz[YEARMONTH %in% trial_period], aes(x = YEARMONTH)) +
  geom_line(aes(y = n_customers, color = Store_type)) +
  geom_ribbon(data = metrics_viz[STORE_NBR == control_store & YEARMONTH %in%
trial_period],
            aes(ymin = n_customers*(1-2*std_customers), ymax =
n_customers*(1+2*std_customers)),
            fill = "grey80", alpha = 0.3) +
  labs(title = paste("Trial Customers vs Control Store", trial_store))
}

write.csv(monthly_metrics, "C:/Users/tracy/Downloads/monthly_metrics.csv",
row.names = FALSE)

```